Trajectory Rewards Are Not Token Credit

Advantage estimator fidelity for long-horizon post-training

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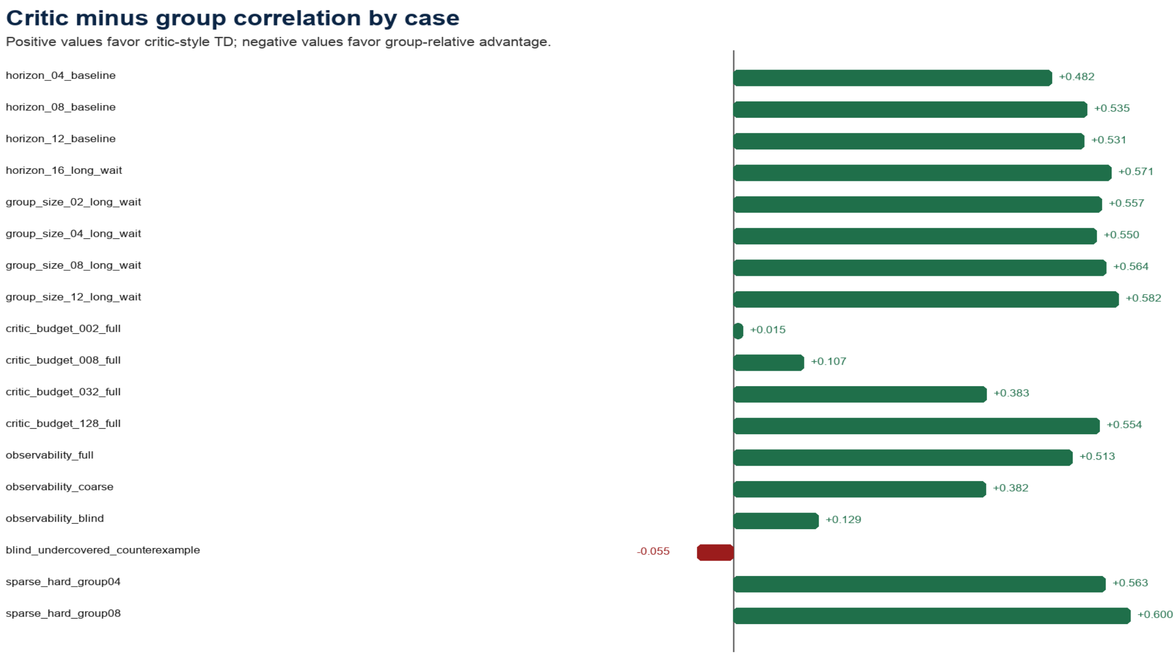
Abstract. We compare PPO-style critic/value-model advantage estimation with GRPO-style group-relative advantages for long-horizon post-training, while situating OPD/OPSD as complementary distillation tools rather than policy-gradient replacements. In a multi-seed toy matrix with known oracle advantages, critic-style TD has the higher mean correlation in 17/18 fixed regimes; a confidence-interval reading gives 16 clear critic-favorable cases, 1 near tie, and 1 clear group-favorable counterexample. Additional audits test anchor-action contrast, coverage-gated credit, and tabular closed-loop training. The conclusion is conditional: PPO-style critics look better when temporal state information is observable and learnable; GRPO remains attractive when reward contrast is reliable and critic coverage/cost is poor.

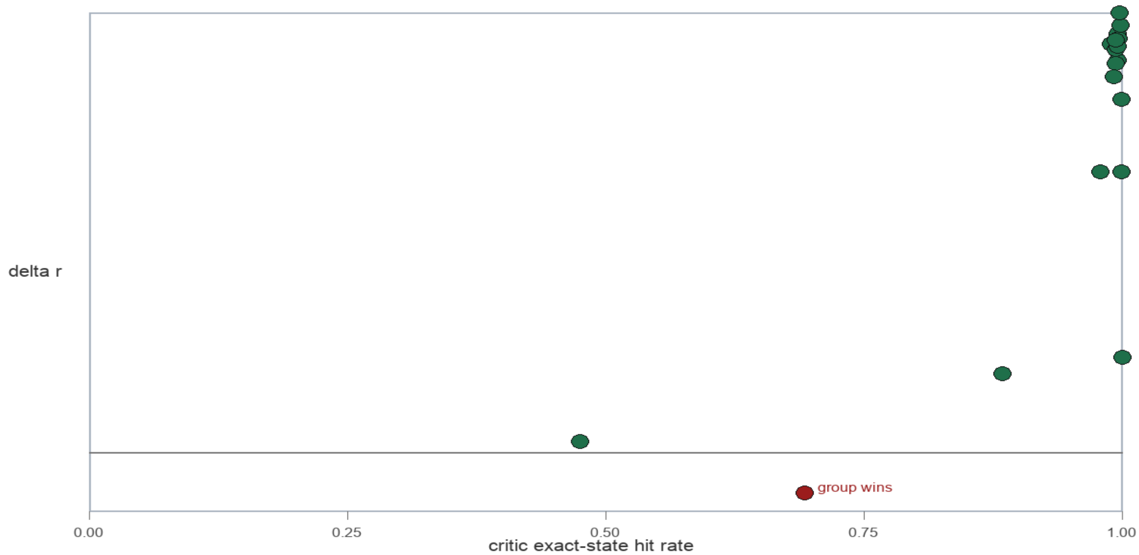
Key Results

Seeds	Cases	Clear critic	Near tie	Clear group	Mean delta r
20	18	16	1	1	0.420



Critic coverage vs estimator advantage

A group-relative counterexample appears when critic state coverage is weak.



Methods

The toy environment has variable-length trajectories with help, harm, and wait actions. A terminal verifier supplies a sparse success reward; known finite-state dynamics provide an oracle state-action advantage. The group-relative estimator normalizes terminal returns within a prompt group and broadcasts the scalar to each token. The critic estimator fits a tabular value model from returns-to-go and computes one-step TD advantages. This measures estimator fidelity, not closed-loop PPO or GRPO learning.

Deep Matrix Results

Case	Axis	Mean	CI read	Group r	Critic r	Delta r	95% CI
horizon_04_baseline	horizon	critic	critic_clear	0.495	0.977	0.482	+/- 0.011
horizon_08_baseline	horizon	critic	critic_clear	0.390	0.925	0.535	+/- 0.015
horizon_12_baseline	horizon	critic	critic_clear	0.335	0.866	0.531	+/- 0.012
horizon_16_long_wait	horizon	critic	critic_clear	0.293	0.865	0.571	+/- 0.013
group_size_02_long_wait	group	critic	critic_clear	0.213	0.771	0.557	+/- 0.022
group_size_04_long_wait	group	critic	critic_clear	0.301	0.851	0.550	+/- 0.014
group_size_08_long_wait	group	critic	critic_clear	0.346	0.910	0.564	+/- 0.011
group_size_12_long_wait	group	critic	critic_clear	0.358	0.940	0.582	+/- 0.007
critic_budget_002_full	budget	critic	near_tie	0.352	0.367	0.015	+/- 0.027
critic_budget_008_full	budget	critic	critic_clear	0.352	0.459	0.107	+/- 0.027
critic_budget_032_full	budget	critic	critic_clear	0.352	0.735	0.383	+/- 0.021
critic_budget_128_full	budget	critic	critic_clear	0.352	0.906	0.554	+/- 0.017
observability_full	observe	critic	critic_clear	0.352	0.864	0.513	+/- 0.020
observability_coarse	observe	critic	critic_clear	0.356	0.738	0.382	+/- 0.015
observability_blind	observe	critic	critic_clear	0.353	0.482	0.129	+/- 0.017
blind_undercovered_counterexample	counter	group	group_clear	0.510	0.455	-0.055	+/- 0.015
sparse_hard_group04	sparse	critic	critic_clear	0.283	0.846	0.563	+/- 0.014
sparse_hard_group08	sparse	critic	critic_clear	0.310	0.910	0.600	+/- 0.008

Interpretation

The evidence supports the PPO-favorable hypothesis in this controlled toy: when progress state is observable and value coverage is adequate, critic-style TD advantages track oracle token credit far better than response-level group scalars.

The counterexample matters. When the critic is blind and undercovered, terminal group-relative outcome information can be more useful than a poor value table. The scientific claim is therefore not that PPO is universally better; it is that value-based temporal information becomes valuable in the long-horizon regime if it can be learned.

Z.ai's GLM-5.2 report is a useful industry case study because it describes compacted, variable-length long-horizon agent rollouts and reports moving from group-wise optimization to critic-based PPO with token-level advantages. It is not independent causal evidence for PPO over GRPO; it is a motivating example that matches the failure mode studied here.

Cost, Safety, and Caveats

Fair comparisons must charge critic memory and training, reward/verifier calls, group sampling, teacher/self-teacher compute, anti-hacking filters, LLM judges, sandboxing, blocked-call handling, and trajectory storage.

Coding-agent RL with pass/fail rewards can incentivize shortcuts. Online tool-call monitoring and invalid-action handling should be treated as part of the training protocol, not merely post-hoc evaluation.

The toy has an oracle and tabular critic. Real systems add neural approximation, KL controllers, stale policies, reward-model error, tool failures, and distributed rollout infrastructure. Stronger GRPO variants and process rewards remain future work.

Selected References

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